FieldBridge: Competitors, Difference, Problem & Economics

- Hackathon research brief · Devlympics Shunya 2026 · Theme: Bridge the Gap · 17 Oct 2026

1. The one-minute version

What it is: FieldBridge takes a short, messy text message from someone in the field, turns it into a clean validated record, replies with "done" or "fix this exact field", and makes sure that sending the same message twice never creates two records.

Honest verdict: REFRAME. The technology works and is demoable in 4 hours, but the idea is not new. Tools like RapidPro already do SMS data collection for free. We should not pitch "first of its kind". We should pitch a narrow, provable promise: **reliable, duplicate-safe, closed-loop messaging**, shown live.

2. What problem are we actually solving?

The simple story

A delivery rider, warehouse worker or event volunteer is out in the field with a basic phone and weak signal. They can send a short text. The office runs on a dashboard that needs clean data: a number that is really a number, a location that actually exists, a status that is one of a fixed list.

Two worlds, two formats. Today a person in the middle usually reads the text, types it into the system, chases missing details, and tells the sender what happened.

Where it breaks (the four real failure cases)

- **Bad data:** the text says "twenty" instead of 20, or the location is missing.
- **Duplicates:** networks retry, so the same message arrives twice and two records get created.
- **Silence:** the sender never learns whether the office received or acted on the message.
- **No connection:** the message can't be sent at the moment it is written.

What we must be upfront about

These failures are logical and widely understood, but **we have not verified how often they hurt real teams**. We have no interviews and no measured numbers, so we do not claim any. In the pitch, call this our hypothesis and show the failures working live instead of quoting statistics.

3. Who are the competitors?

Competitor	What it does	Reported price	How close to us
RapidPro / TextIt	Free open-source platform (built with UNICEF) to send and receive data by SMS, voice and chat apps, with visual flows and an API. TextIt is the commercial hosted version.	RapidPro free (self-host); TextIt paid	**Closest competitor.** Already does SMS data collection and automated replies.
FrontlineSMS and similar SMS gateways	Match keywords in incoming texts and route them.	Varies	Handles receiving and routing, not much validation or case tracking. Product status needs checking.
KoboToolbox	Offline mobile forms for field surveys, widely used in humanitarian work.	Free for non-profits; paid plans for companies reported from about $159/month	Solves offline forms on smartphones, not text-message input.
ODK (Open Data Kit)	Open-source offline data collection.	Free to self-host; ODK Cloud reported from about $199/month	Same as Kobo: strong at forms, not SMS.
SurveyCTO	Paid, polished field data collection.	Reported from about $349/month	Same category as Kobo and ODK.

Competitor	What it does	Reported price	How close to us
Twilio, MSG91, Fast2SMS	Pipes that carry SMS. They send and receive but don't define rules.	Per message (see section 5)	We would sit on top of one of these, so they are partners, not rivals.
WhatsApp groups and spreadsheets	What most small teams use today.	Almost free	The real "competitor" for small teams: no validation, no dedupe, manual copying.
ChatGPT or any LLM	Can turn a sentence into JSON.	Subscription or API	Can parse, but has no memory of past messages, no clock and no state.

Prices come from third-party blog posts, some written by competing vendors. Check each official pricing page before putting a number on a slide.

4. How are we different?

Where we overlap (say this openly)

Offline forms, SMS collection, validation and automated replies already exist. If a judge says "RapidPro can do this", the correct answer is "yes, partly, and here is what we focus on".

Our narrow, defensible focus

We are not a form builder or a general bot platform. We are a small, strict **message protocol with guarantees**:

- **Message ID plus Case ID.** Every text has a unique ID, so retries are recognised and ignored. A new message with the same Case ID updates the same case instead of creating a new one.
- **Exact correction replies.** Not "invalid input" but "QTY must be a number, LOC is missing". The failed attempt stays in the history.

- **Closed loop.** The case only finishes after: sent, validated, acknowledged by the office, confirmed by the sender. Each step is a tracked state.
- **Stretch goals, if time allows:** a timer that chases a sender who hasn't confirmed, and a tamper-evident audit log where each entry carries the hash of the one before it.

We have **not verified** whether RapidPro handles duplicate messages and state tracking out of the box. Check this before claiming that it doesn't.

One-line pitch

> FieldBridge makes short field messages safe to act on: validated, never duplicated, and always answered.

5. Price economics

What the transport costs

For India, business SMS runs through a mandatory registration system (DLT): you register your sender name and your **exact message templates** before anything is delivered. Practical effect: our acknowledgement and correction replies must be written as pre-approved templates with blanks, not freely generated text.

Item	Reported cost
Business SMS via Indian providers at volume	about ₹0.12 to ₹0.30 per SMS, before GST
Twilio for India SMS	reported at roughly ₹0.45 per message equivalent, often several times the Indian providers (a competing vendor's claim, so verify)
Per-SMS operator scrubbing fee	a few paise
Self-hosted ODK, Kobo (non-profit) or RapidPro	software free, you pay hosting and your own time

A worked example (illustrative, not measured)

Suppose a team receives **500 reports a day**, about 15,000 a month. Each gets one acknowledgement, and roughly 10% need a correction reply, so about 16,500 outgoing messages.

- At ₹0.15 each: about **₹2,500 a month**
- At ₹0.30 each: about **₹5,000 a month**

Compare with manual handling. If re-typing and replying takes about 1 minute per report, that is roughly 250 hours a month of someone's time. That figure is an assumption, but it shows why paying a few thousand rupees to remove the middleman could make sense. We would need real teams to confirm it.

Possible business models (hypotheses)

- **Pay per message:** small markup on the SMS cost.
- **Flat monthly tier** by number of field users.
- **Open core:** the protocol is free, hosting and support are paid.

Cost of our hackathon prototype

₹0. Transport is simulated (a fake phone screen plus a connection toggle) and runs on free hosting. We must say clearly that real SMS integration is future work.

6. What we know vs what we are guessing

Status	Statement
Verified	RapidPro is a free, open-source SMS data-collection platform. Kobo and ODK handle offline field forms. India requires registered SMS templates.
Reported, needs checking	Competitor prices; Twilio being 2 to 4 times more expensive in India.
Hypothesis	Teams lose time and data to bad, duplicate or unanswered messages. A narrow tool would be worth paying for.

Status	Statement
Prototype only	Offline mode is a simulated outbox. No real SMS is sent.
Not claimed	Any measured impact, time saved or errors reduced.

7. How to pitch it

Frame: "Bridge the gap between a one-line text and a system that needs clean records."

Demo order: send a valid message, see the case appear; turn the connection off, send another, see it wait, turn it on, see it sync; resend the same message, see one case; send a broken message, see the exact correction; finish the loop with acknowledge and confirm; end on the audit history.

Toughest questions, short answers

- **"Isn't this just a parser?"** Parsing is step one. Duplicate-safe retries, case state, corrections and confirmation are the product, and we show them live.
- **"Why not RapidPro or Kobo?"** They are strong tools. We are testing a narrow, ready-made protocol focused on reliability and closing the loop, and we aren't claiming to replace them.
- **"Why not ChatGPT?"** It can read a message, but it can't remember it saw it yesterday or chase a sender who went quiet. Our backend can.
- **"What impact did you measure?"** None. This is a technical prototype and we say so.

8. Sources used

- UNICEF and IADB pages describing RapidPro and TextIt
- Engineering for Change entry on RapidPro
- Clappia comparison of KoboToolbox alternatives (ODK Cloud and SurveyCTO prices)
- Kolonell blog posts on offline data collection costs (Kobo paid-plan price)
- Message Central, RichAutomate and Botble SMS provider guides (India SMS costs and DLT rules)

- A 2015 University of Washington course deck on SMS for development (keyword-based tools)
- Official sites to check before the final pitch: kobotoolbox.org/pricing, docs.getodk.org, rapidpro.io, twilio.com/docs